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B737 FLEET MAGAZINE

An exclusive fleet magazine for EAL 737 pilots.

Fleet News · Deeper look into B737 incidents and accidents · Breakthroughs in flight · Notices · System reviews

NOTE

This magazine is for information purposes only. **USE COMPANY AND APPROVED MANUALS AS PRIMARY REFERENCE**.

Thank you for taking time to read The EAL 737 Fleet MAGAZINE. All readers are encouraged to submit their questions and suggestions, regarding this magazine to israelya@ethiopianairlines.com

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December 2025

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MESSAGE FROM CHIEF PILOT

Greetings to you all,

Over the past three months, our fleet and flight operations have experienced a period of both challenge and meaningful progress. Despite the demanding nature of the work, I am pleased to reflect on the developments that have unfolded since the last newsletter and to highlight the initiatives that continue to strengthen our operational safety, efficiency, and crew cohesion.

To begin with, we successfully received a new Boeing 737 MAX aircraft last month, though it was redelivered to ASKY the ferry flights were conducted by our pilots. Alongside this addition, fleet-wide performance indicators have shown encouraging trends. **Notably, hard landings and deep landings have decreased, demonstrating improved discipline and situational awareness during the critical phases of flight.**

While no significant incidents were recorded during this period, the highest FDM-triggered event remains a low threshold crossing height. Some approaches have been flown below the published profile, possibly due to the expectation of a reduced ground roll.

This serves as an important reminder that maintaining proper vertical guidance is essential not only for procedural compliance but also for operational safety.

In the same spirit of continuous improvement, **we have introduced a new one-step-down climb power selection procedure.** Based on guidance from the engine manufacturer, this modification allows us to extend the onwing life of our engines, ultimately contributing to significant cost savings for the airline. Under this new procedure, crews are required to select a climb thrust level one step below the FMC's default recommendation. For example, when the FMC suggests CLIMB 1 under current environmental conditions, pilots must manually select CLIMB 2 instead. **We are proud to be only the second airline globally to implement this forward thinking practice**, and we look forward to the longterm benefits it is expected to yield.

Continued on next page ...

As we prepare for the **upcoming IOSA audit, scheduled for January 26–30, 2026**, I would like to underscore the importance of this event. IOSA accreditation is vital for safeguarding our membership in IATA. Losing IATA membership would have wide-ranging and debilitating effects on the airline—impacting everything from ticketing and interline agreements to cooperation with other carriers. While the audit occurs every two years, the standards it examines must be embedded in our daily operations.

This is not an exercise reserved for audit season; rather, **it reflects a culture of safety and professionalism that we must uphold yearround**. Each one of us contributes to this standard by adhering strictly to SOPs, complying fully with regulations, and maintaining unwavering safety awareness.

The Senior Captain Initiative, which was elaborated upon in the previous newsletter, also continues to advance effectively. Of particular note is the **First Officer Development Program**, launched two months ago, designed to strengthen CRM competencies across the FO community.

These improvements directly enhance fleet safety and cockpit coordination. Looking ahead, we are preparing to include Q400 First Officers in the upcoming online monthly development sessions so that all segments of our operation benefit from consistent professional growth.

Additionally, **a new focal point has been established to improve communication and issue resolution with outstation managers**. This role addresses key challenges such as transportation, accommodation, and other logistical concerns experienced at outstations, ensuring more responsive and effective support for our crews.

Looking forward to the next quarter, the fleet is expected to expand with the delivery of **five more Boeing 737 aircraft**. Two of these will join the ASKY fleet, while one is planned for redelivery to Congo Airlines. Moreover, a new vacancy for the Senior Captain position will soon be announced. Many have found fulfillment and purpose in contributing to the airline through such leadership roles, and this new posting will provide another group of senior captains the opportunity to do the same.

Finally, I would like to extend heartfelt congratulations to colleagues who have recently married or welcomed new additions to their families. We also offer our deepest condolences to those who have lost loved ones during this period. As always, we stand together as one team—supporting each other through both joyous and difficult moments.

Lastly I wouldn't want to forget the diligence demonstrated by our FOs in the last two months. They went extra mile in supporting the operation. Keep that up.

Safe Flight,
Capt. NATHAN ELIAS
Chief Pilot – B737&B767

ETHIOPIAN AIRLINES GROUP SAFETY POLICY

Ethiopian is committed in developing, implementing, and continuously improving strategies and processes to ensure that all operations achieve the highest level of safety performance. Ethiopian's commitment is to:

- Provide the necessary resources to implement this safety policy.
- Establish an organizational safety culture that prioritizes safe practices, foster effective safety reporting, and communication.
- Ensure the safety policy is communicated throughout the organization and reviewed not less than once every two years.
- Clearly define individual accountabilities and responsibilities for the organization's safety performance.
- Ensure effective management of safety and security risks to aircraft operations.
- Establish and maintain proactive as well as reactive hazard identification and risk management processes, to mitigate safety risks and ensure continuous improvement.
- Conduct regular analysis of malfunctions or undesirable operational results.
- Follow-up of corrective actions and their effectiveness in improving operational performance. Ensure all staff involved in Safety Management System are adequately trained.
- Ensure that punitive actions are not taken against any employee who discloses safety concerns through safety reporting system unless this disclosure indicates, beyond reasonable doubt, an illegal act, gross negligence, or willful disregard of regulations or procedures.
- Comply with legislative and regulatory requirements as well as with Ethiopian Airline's standards.
- Ensure that sufficient skilled and trained human resources are available to implement safety strategy and processes.
- Continuously improve safety performance through the monitoring, measurement, and review of safety objectives and targets by senior management.
- Establish and measure safety performance against realistic safety performance indicators and targets.
- Ensure the promotion of safety and security awareness to all employees and stakeholders. Ensure all outsourced products and services that impact safety performance meet applicable Ethiopian Airlines' standards.

SAFETY IS OUR FIRST PRIORITY!

Across the fleet, a number of practical updates and initiatives have been introduced over recent months, each aimed at supporting safer operations, clearer communication, and more efficient day-to-day flying. From digital tools that improve access to operational information, to targeted development programs and important manual revisions, these updates reflect a continued focus on learning, standardization, and continuous improvement. The following highlights summarize the key developments since the last issue.

1.Telegram Bots

Over the past months, several practical digital tools and initiatives have been introduced to improve day-to-day communication, situational awareness, and operational coordination across the fleet.

Dedicated Telegram groups and bots are now in active use, including a maintenance and safety channel where aircraft status updates, safety information, and operational news are shared in real time.

A maintenance bot (@ETMELCDL_Bot) allows pilots to quickly retrieve deferred item information by entering an aircraft registration, while a scheduling bot (@et_scheduling_team_bot) provides a direct and structured way for pilots to flag roster concerns at the end of the current month and the start of the next, improving transparency and responsiveness.

2.First Officer Development Program

In parallel, the First Officer Development Program continues to gain momentum as a valuable experience-sharing platform.

The program brings together First Officers and some of the airline's most respected and experienced captains for scenario-based discussions and lessons learned from years of line operations. Sessions are conducted on Microsoft Teams, with meeting links distributed via email and fleet Telegram groups by Senior Captains, ensuring wide and easy access for all participants.

3.SOP AND FOM REVISIONS

On the documentation front, SOP Revision 9 and FOM Revision 25A have now been published.

Key **SOP updates** include changes to standard callouts, particularly those related to command speed during approach and Engine failure Procedures during climb.

FOM updates include revisions in Chapter 1 covering organizational structure, uniform policy, the use of non-standard adapters, and baggage handling procedures. Chapter 2 reflects updates to stabilized approach criteria and recent bulletins, while Chapter 3 introduces a generic engine failure procedure. Pilots are encouraged to review the revised manuals in full to familiarize themselves with all changes.

4.CLIMB POWER SELECTION

As part of ongoing efforts to balance operational efficiency with long-term reliability, a revised climb power selection procedure has been introduced across the entire Boeing 737 fleet, effective immediately.

Under this change, crews are now requested to select one step lower than the FMC-provided climb power setting during climb.

For example, if the FMC displays CLIMB, crews should manually select CLIMB 1, and if CLIMB 1 is provided, CLIMB 2 should be selected.

This adjustment follows extensive data analysis by the engine manufacturer, which has demonstrated that operating at a reduced climb thrust profile significantly extends engine life while delivering meaningful cost savings for the airline—without compromising safety or performance.

As always, operational requirements take precedence. If ATC instructions, terrain, weather, or performance considerations require full climb thrust, crews should not hesitate to select the appropriate climb power as needed.

VOLCANIC ASH

MANAGING THE EFFECTS

A commercial airplane encounter with volcanic ash can threaten safety of flight because of resulting conditions that range from windshield pitting to loss of thrust in all engines. Unlike ordinary dust or smoke, volcanic ash is made up of tiny, sharp fragments of rock, minerals, and volcanic glass formed during explosive eruptions. These particles can rise to cruising altitudes used by commercial jets and spread thousands of kilometers from the volcano, often invisible to onboard radar.

Flight into a volcanic ash cloud is considered unsafe and should be avoided. Severe adverse effects includes engine flame-out, engine overheating, clogging of pitot-static probes, abrasion of external/internal components, etc

1. Results of past events involving volcanic ash

Over 80 aircraft have reported to have flown into volcanic ash cloud between 1980 and 2000, with consequences ranging from increased wear of engines to simultaneous power loss in all engines.

These are some of the most significant volcanic ash events on record:

<u>1982</u>: British Airways Flight 009 (Mount Galunggung, Indonesia): A 747, all four engines failed, crew restarted engines and landed safely, suffering significant engine and surface damage.

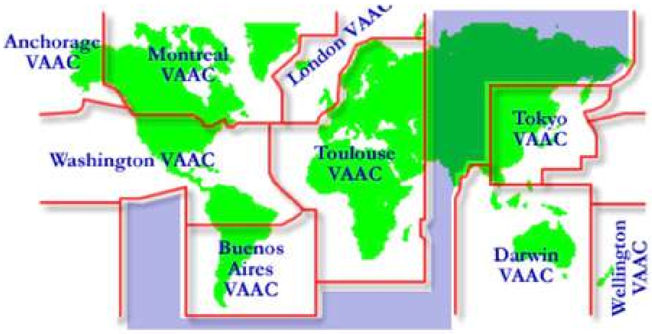
<u>1989</u>: KLM Flight 867 (Mount Redoubt, Alaska): Another 747 , lost all engines, successfully restarted them at lower altitudes and land in Anchorage, with extensive engine damage.

<u>1991</u>: Mount Pinatubo (Philippines): Caused numerous incidents, leading one airline to ground planes in Manila for days and highlighting the danger of ash.

<u>2010</u>: Eyjafjallajökull (Iceland): The most significant disruption, closing European airspace for days, stranding millions, and prompting major changes in how ash is managed.

2. Resources available to help avoid ash encounters

Alert messages (volcanic ash SIGMET) are issued by a Meteorological Watch Office (MWO) for its area of responsibility. Nine Volcanic Ash Advisory Centers (VAAC) have been designated by international organizations to provide an expert advice to MWO regarding the location and expected movement of volcanic ash clouds.



3.Specific flight crew actions required in response to encounters

Despite ongoing avoidance efforts, operators can still experience volcanic ash encounters. Guidance on the operational issues surrounding volcanic ash is divided into three aspects:

- Avoidance
- Recognition, and
- Procedures.

Avoidance- Preventing flight into potential ash environments requires planning in these areas:

- Dispatch needs to provide flight crews with information about volcanic events, such as potentially eruptive volcanoes and known ash sightings that could affect a particular route.
- Dispatch also needs to identify alternate routes to help flight crews avoid airspace containing volcanic ash.
- Flight crews should stay upwind of volcanic ash and dust.
- Flight crews should note that airborne weather radar is ineffective for distinguishing ash and small dust particles.

Recognition

Indicators that an airplane is penetrating volcanic ash are related to **odor, haze, changing engine conditions, airspeed, pressurization, and static discharges.**

- Odor. When encountering a volcanic ash cloud, flight crews usually notice smoky or acrid odor that can smell like electrical smoke, burned dust, or sulfur.
- Haze. Most flight crews, as well as cabin crew or passengers, see a haze develop within the airplane. Dust can settle on surfaces.
- Changing engine conditions. Surging, torching from the tailpipe, and flameouts can occur. Engine temperatures can change unexpectedly, and a white glow can appear at the engine inlet.
- Airspeed. If volcanic ash fouls the pitot tube, the indicated airspeed can decrease or fluctuate erratically.
- Pressurization. Cabin pressure can change, including possible loss of cabin pressurization.
- Static discharges. A phenomenon similar to St. Elmo's fire or glow can occur. In these instances, blue-colored sparks can appear to flow up the outside of the windshield or a white glow can appear at the leading edges of the wings or at the front of the engine inlets.

Procedures

FOM 7.13.3 states: When encountering visible ash cloud the following generic procedures are recommended; however, flight crew should reference the specific Aircraft Operating Manual/Emergency Checklist for detailed guidance:

- Turn on continuing ignition.
- Declare an emergency.(MAY-DAY, MAY-DAY, MAY-DAY)
- Do not climb in order to over-fly the ash cloud.

- Reduce power to idle in order to provide additional engine stall margin and lower turbine temperature.
- Try to escape the ash cloud by turning 180 degrees and descending (if terrain clearance permits).
- Monitor altitude versus airspeed.
- Turn on all accessory air bleeds including all air conditioning packs, nacelles, and wing anti-ice. This will provide an additional engine stall margin by reducing engine pressure.

GPS SPOOFING

AN EMERGING THREAT TO MODERN NAVIGATION

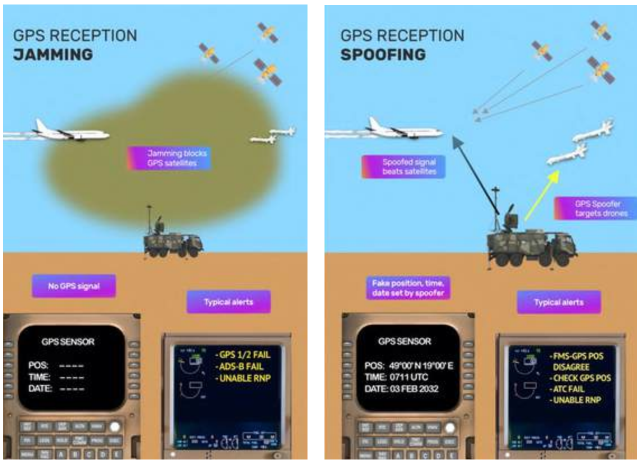
1.0 Operational Considerations

Operational Considerations for Boeing 737 Crews Over the last few years, GPS spoofing has evolved from a localized anomaly into a recurring operational challenge affecting commercial aviation worldwide. According to the OPS GROUP GPS Spoofing Final Report, incidents have been reported across multiple regions, with increasing frequency and sophistication.

Unlike GPS jamming, which simply denies signal reception, spoofing involves the transmission of false GPS signals that appear valid to the aircraft, potentially misleading onboard navigation systems without immediate warning .

For Boeing 737 NG and MAX crews, the implications are particularly significant due to the aircraft's high reliance on GNSS inputs for navigation, flight guidance, and certain flight deck indications. Spoofing events have resulted in false position shifts, incorrect groundspeed and track information, map display displacement, and unexpected navigation mode changes.

In some cases, aircraft have appeared on the Navigation Display to be offset miles from their actual position, despite stable flight parameters and no apparent system faults .



2.0 How GPS Spoofing Manifests

On the 737, GPS inputs are integrated into the FMC, IRS updating logic, and map displays. When spoofed signals are accepted, crews may observe symptoms such as **sudden map shifts, loss of RNP capability, "GPS-L INVALID and GPS-R INVALID", "UNABLE REQD NAV PERF-RNP","VERIFY POS FMC" or "TERR POS"** messages, and disagreement between raw data and FMC-derived information.

Reports highlight that many spoofing encounters were initially detected by vigilant crews comparing FMC position with raw radio navigation aids, visual references, or expected waypoint sequencing. This reinforces a fundamental principle : **automation should always be monitored and verified, not assumed correct!**

3.0 Operational Risks and Flight Phase Sensitivity

GPS spoofing poses the highest risk during terminal operations, RNP approaches, and operations in constrained airspace where lateral accuracy is critical. For crews operating RNP AR or RNAV (GNSS) approaches, false position data can lead to incorrect lateral guidance, unstable approaches, or last-minute go-arounds.

During cruise, the impact is often less immediate but may still result in incorrect fuel predictions, route deviations, or loss of required navigation performance.

Reports emphasize that reliance on GPS-based navigation should always be balanced with situational awareness and alternative navigation methods, especially in regions known for GNSS interference.

4.0 Crew Actions and Best Practices

Early recognition and disciplined response is strongly advocated.

For B737 crews, recommended actions include

- Tuning and cross checking FMC position from available ground based navigation aids
- Deselecting unreliable FMC guidance if necessary as per the supplementary procedures
- Cross-checking IRS position against navigation aids
- Advise ATC promptly.
- Crews are also encouraged to file reports following new suspected spoofing encounters. These reports contribute to industry awareness, improve threat mapping, and support mitigation strategies at both operator and regulatory levels.

5.0 Looking Ahead

GPS spoofing is not a theoretical threat; it is a present operational reality. While technological counter measures continue to evolve, the most effective mitigation remains an informed, alert, and well-trained flight crew.

For Boeing 737 NG and MAX operations, reinforcing raw data proficiency, understanding system behavior, and maintaining healthy skepticism of automation outputs are essential to safely navigating this growing challenge.

Pilots are encouraged to familiarize themselves with company guidance, Boeing recommendations, and other reports to enhance awareness and preparedness.

RUNWAY INCURSION

PREVENTION METHODS

1.0 What is a Runway Incursion?

Any occurrence at an airport movement area involving: An aircraft, Vehicle, Person, Animals or object on the ground that creates a collision hazard or results in loss of separation with an aircraft, taxiing, intending to take off, landing, or intending to land.

2.0 FLIGHT CREW PROCEDURES

The potential for runway incursion, incident and accident can be reduced through adequate planning, coordination and communication. The following guidelines are intended to help flight crews cope more effectively with current airport conditions during taxi operations.

2.1.Planning

Thorough planning for taxi operation is essential for safe operation. Flight crews should give as much attention to the planning of the airport surface movement portion of the flight as they give to the planning of the other phases of flight.

Planning should be done in two main phases:

1. Anticipate airport surface movements by doing pre-taxi or pre-landing planning based on the weather report and on previous experience at the airport.

1. Once taxi instructions are received, the pre-landing or pre-taxi plan should be reviewed and updated as necessary. It is essential that the updated plan is understood by all flight crew members.

<u>CAUTION</u>

A potential pitfall of pre-taxi and pre-landing planning is setting expectations and then receiving different instructions from the ATC. Flight crews need to ensure that they follow the clearance or instruction that are actually received, and not the one the flight crew expected to receive.

The following guidance should be used to conduct a briefing of all crew members

- How familiar are the crew-members with the airport?
- Has any member of the crew flown out of or into the airport recently?
- Might there be some changes made at the airport recently?

REMEMBER

- To review the latest NOTAM for both the departure and arrival airport for information concerning construction and Arrival airport for information concerning construction or taxiway/runway closures.

- Take some time and study the airport layout. An airport diagram must be readily available for use by the pilots.

- Pay special attention to any unique or complex intersections along the taxi route(Hot Spots). Flight crews should identify critical times and location on the taxi route (transitioning trough complex intersections, crossing intervening runways,

- The verbal coordination between the PIC and the FO will be important to ensure correct aircraft navigation and crew orientation.
- The flight crew should plan the timing and execution of checklists and company communication (IOCC) at the appropriate times and location.
- All intersection should be considered as the airport "HOT SPOTS"‼
- If at all possible, during low visibility operation flight crews should only conduct before-takeoff checklist when aircraft is stopped.

2.2.SITUATIONAL AWARENESS

When conducting taxi operations flight crew need to be aware of their situation as it relate to the other aircraft operations going on around them as well as other vehicles moving on the airport.

The flight crew should know the aircraft's precise location on the airport. Especially when the visibility is poor. It is important for the flight crew to understand and follow ATC instructions and clearance.

Flight crews should use a "continuous loop" progress for actively monitoring and updating their progress and location during taxi. Verbal sharing of relevant information between the crew members is essential.

Situational awareness is enhanced by monitoring ATC instruction/clearances issued to other aircraft.

NOTE

- Prior to entering or crossing any runway, scan the full length of the runway (RVR permitting) including approach areas.
- Verbally confirm scan results with each other and aircraft movement should be stopped if there is any difference or confusion on part of any flight crew member about the result of the scan.

<u>CAUTION</u>

Do not stop on the runway. If possible, taxi off the runway and then initiate communications with the ATC to regain orientation. Be specially vigilant when instructed to taxi into position and hold, particularly at night or during periods of reduced visibility.

Do not remain in position or hold on the departure runway for an extended period without direct communication from ATC. If any flight crew member is uncertain about any ATC instruction or clearance, query ATC immediately.

All flight crewmembers must have a common understanding of ATC's instructions and expectations regarding where the aircraft is to stop and must be able to identify the appropriate hold points. When in doubt, immediately advise ATC about the ability to comply with any of their instructions.

<u>CAUTION</u>

- Unless otherwise instructed by ATC, clear of the landing runway even if that requires you to cross or enter a taxiway/ramp area.
- Never enter a runway without specific authorization. When in doubt, confirm with ATC.
- For inbound aircraft,scanning full length of the runway including approach and departure end of the runway before crossing.

2.3. USE OF WRITTEN TAXI INSTRUCTIONS

At many airports, taxi instruction can be very complex, involving numerous turns and transitions as well as runway crossing and holding instructions.

During the part of the surface operation, pilots are very busy with a variety of cockpit duties and responsibilities that compete for their attention. Writing down taxi instruction especially complex instructions, can reduce a pilot's vulnerability to forgetting part of a complex instruction and can be used to support airport surface operations as follows:

1. For use as reference for reading back the instruction to ATC.
2. For crewmembers coordination on the assigned runway and taxi route.
3. For short pre taxi or pre landing briefing on the pending airport surface operation.
4. As a means of reconfirming the taxi route and any restriction at any time during the airport surface operation.

2.4. INTRA-FLIGHT DECK/COCKPIT VERBAL COORDINATION

It is essential that the flight crew correctly understand and agree on all ATC ground movement instructions.Any misunderstanding or disagreement should be resolved before taxiing the aircraft.

It is the verbal aspect of this coordination that is most significant. It is not correct to assume that all flight crew members have heard and understood instructions correctly. Any persistent disagreement or uncertainty among crew members should be resolved by contacting ATC for clarification.

This verbal coordination /agreement should be accomplished:

- When ATC issues taxi instruction for departure, the flight crew should refer to the airport diagram, coordinate verbally and agree on assigned runway and taxi route, including any instruction to hold short of or cross an intersecting runway.
- When ATC issues landing instructions, the flight crew should coordinate verbally and agree on the runway assigned by ATC, as well as any restrictions, such as hold short points of an intersecting runway after landing.
- When approaching an intersecting runway,the flight crew should coordinate verbally in order to identify the runway. They should also verbally review the ATC instructions as to whether they are to hold short of or cross the runway.
- Before entering a runway for takeoff, the flightcrew should verbally coordinate to ensure correct identification of the runway and receipt of the proper ATC clearance to use it. Similar verification should be performed during approach and landing.
- When it becomes necessary for a flight crew member to stop monitoring any ATC frequency, he/she should advise the other crewmember(s) when stopping and resuming the frequency monitor.
- Any information or instructions received or transmitted during that flight crewmember's absence from the ATC frequency should be briefed and reviewed upon his/her return.

2.5 ATC/FLIGHT CREW COMMUNICATION

The primary way the flight crew and ATC communicate is by voice. The safety and efficiency of taxi operations depends on this communication loop.

Controllers use standard phraseology and require readbacks and other responses from the flight crew in order to ensure that clearances and instructions are understood.

In order to complete the communication loop, the controllers must also clearly understand the flightcrew's read back and other responses. The flight crew can help enhance the controller's understanding by responding appropriately and using standard phraseology.

Some of the most important guidelines that contribute to clear and accurate communications :

1. Maintain a sterile cockpit. Flight crewmembers must be able to focus on their duties without being distracted by non-flight related matters (eating meal, reading non-related material or engaging in non-essential conversation etc)
2. Use standard ATC phraseology at all times in order to facilitate clear and concise ATC/flight crew communications
3. Focus on what ATC is instructing. Do not perform any non-essential tasks while communicating with ATC.
4. Readback all hold short and runway crossing instructions and clearances, including runway designator. Air Traffic controllers are required to obtain from the pilot a readback of all runway hold short from the pilot a readback of all runway hold short instructions.
5. Read back all takeoff and landing clearances, including the runway designator.
6. Use all resources available, including heading indicators,airport signs, marking and lighting, to keep the aircraft on its assigned taxi route.

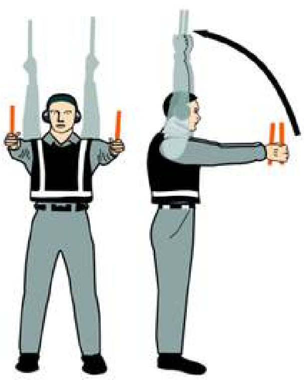
MARSHALLING

Normal and Emergency Signals

1.SIGNALS BY MARSHALLER TO PILOT

1.WINGWALKER/GUIDE
Raise right hand above head level with wand pointing up; move left-hand wand pointing down toward body.

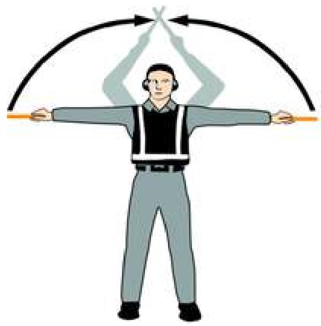
NOTE: This signal provides an indication by a person positioned at the aircraft wing tip, to the pilot/marshaller/push-back operator, that the aircraft movement on/off a parking position would be unobstructed.



2.IDENTIFY GATE
Raise fully extended arms straight above the head with wands pointing up.

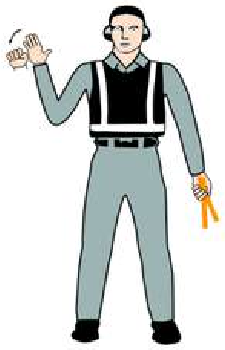
3.PROCEED TO THE NEXT SIGNALMAN OR AS DIRECTED BY TOWER/GROUND CONTROL
Point both arms upward; move and extend arms outward to sides of body and point with wands to the direction of next signalman or taxi area.

4.STRAIGHT AHEAD
Bend extended arms at elbows and move wands up and down from chest height to head.



5.NORMAL STOP
Fully extend arms and wands at a 90-degree angle to sides and slowly move to above head until wands cross.

6.EMERGENCY STOP
Abruptly extend arms and wands to the top of the head, crossing wands.



7.SET BRAKES
Raise hand just above shoulder height with an open palm. Ensuring eye contact with the flight crew, close hand into a fist. Do not move until receipt of "thumbs up" acknowledgment from the flight crew.

8.RELEASE BRAKES
Raise hand just above shoulder height with the hand closed in a fist. Ensuring eye contact with the flight crew, open palm. Do not move until receipt of "thumbs up" acknowledgment from the flight crew.

9.CHOCKS INSERTED
With arms and wands fully extended above head, move wands inward in a "jabbing" motion until wands touch. Ensure acknowledgment is received from the flight crew.

10.CHOCKS REMOVED
With arms and wands fully extended above head, move wands inward in a "jabbing" motion until wands touch. Ensure acknowledgment is received from the flight crew.

11.START ENGINES
Raise right arm to head level with wand pointing up and start a circular motion with the hand; at the same time, with left arm raised above head level, point to the engine to be started.

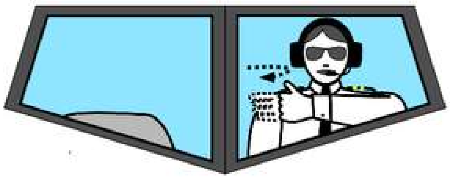
12.CUT ENGINES
Extend arm with wand forward of the body at shoulder level; move hand and wand to top of left shoulder and draw wand to top of the right shoulder in a slicing motion across the throat.

13.SLOW DOWN
Move extended arms downwards in a "patting" gesture, moving wands up and down from the waist to knees.

14.HOLD POSITION/STANDBY
Fully extend arms and wands downwards at a 45-degree angle to sides. Hold position until the aircraft is clear for the next maneuver.

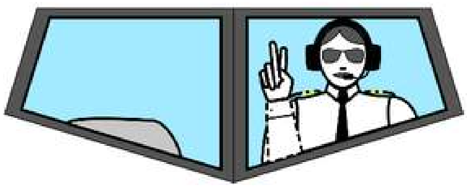
2. PILOT TO MARSHALLER

1.BRAKES ENGAGED
Raise arm and hand with fingers extended horizontally in front of the face, then clench fist.



2.BRAKES RELEASED
Raise arm with fist clenched horizontally in front of the face, then extend fingers.

3.REMOVE CHOCKS
Hands crossed in front of the face, palms facing outwards, move arms outwards.



4.READY TO START ENGINES INDICATED
Raise the number of fingers on one hand indicating the number of the engine to be started.

3. STANDARD EMERGENCY HAND SIGNALS

The following hand signals are established as the minimum required for emergency communication between the aircraft rescue and firefighting (ARFF) incident commander/ARFF firefighters and the cockpit and/or cabin crews of the incident aircraft. ARFF emergency hand signals should be given from the left front side of the aircraft for the flight crew.

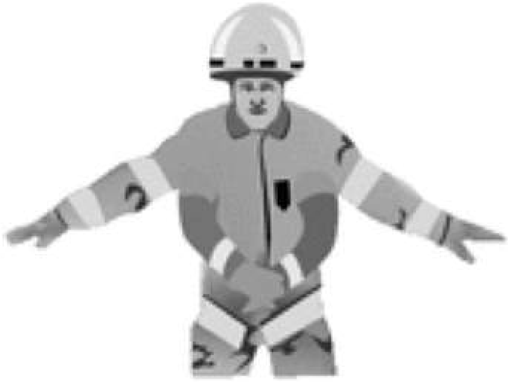
1.RECOMMEND EVACUATION
Arm extended from body and held horizontal with hand upraised at eye level. Execute beckoning arm motion angled backward. Non-beckoning arm held against body.

Night — same with wands.

2.RECOMMENDED STOP
Recommend evacuation in progress be halted. Stop aircraft movement or other activity in progress.

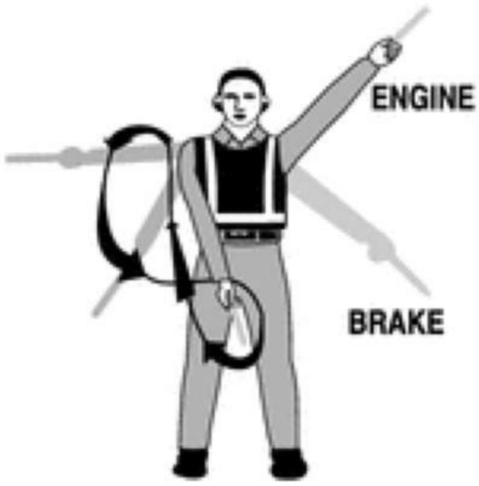
Arms in front of head, crossed at wrists.

Night — same with wands.



3.EMERGENCY CONTAINED
No outside evidence of dangerous conditions or “all-clear” Arms extended outward and down at a 45-degree angle. Arms moved inward below waistline simultaneously until wrists crossed, then extended outward to starting position (umpire's “safe” signal).

Night — same with wands.



4.FIRE
Move right hand in a “fanning” motion from shoulder to knee, while at the same time pointing with left hand to area of fire.

Night — same with wands.

FUELING & DEFUELING

PRECAUTIONS AND STANDARD PROCEDURES

1.0 INTRODUCTION

Aircraft refuelling and defuelling are accompanied by attendant hazards which must be managed sufficiently for their mitigation to acceptable levels.

The aerodrome operator, the aircraft operator and the fuel supplier each has responsibilities in respect of the safety measures to be taken during fuelling operations.

FLIGHT CREW SHALL

- Confirm presence of fire truck (applies only if it is a country or aerodrome requirement).
- Confirm presence of a FT who shall maintain two-way communication either by the aircraft intercom system or other suitable means .
- Notify Cabin TL commencement of fueling
- **Notify to the fire service the seat number of passenger on a stretcher.**
- Turn OFF seat belt signs. (Turn on when hose disconnected & panel closed)

STANDARD PRECAUTIONS FOR FT & FC DURING FUELING/DEFUELING OPERATIONS

- The PIC and fueling technician must confirm the presence of the fire truck unless the state aerodrome requirement states otherwise.
- Fueling technicians must know the most expeditious means to call fire brigade.
- Approved flameproof electrical equipment in the vicinity of the aircraft may be left on if essential, other equipment should be switched off.
- Personnel movement within the safety area shall be reduced to a minimum; however, under no circumstances shall:

1. Passengers remain or walk in the safety area.
2. Electrical connection be stopped or established.
3. Photographic flash bulbs and electronic flash equipment be used within 3m of the filling and venting points on the aircraft or fueling equipment.
4. Anybody smokes in the area.

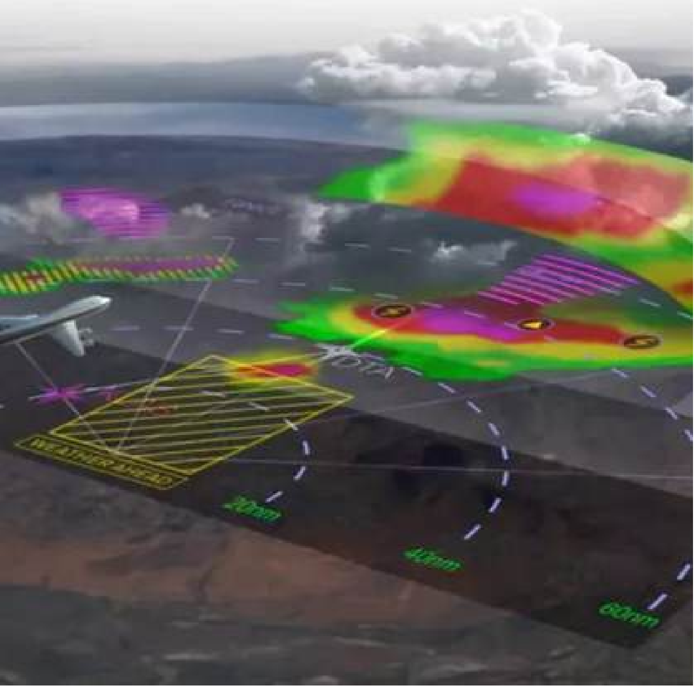
- In the absence of anti-static additive, the fueling rate must be half of the normal rate.
- No fueling operations shall take place during storms with high risk of static discharges.
- **Do not operate HF radios during refueling/defueling operations.**
- **Do not operate the weather radar within 50 feet of fueling/defueling operations or fuel spills.**
- Avoid switching on or off electrical or electronic circuits or carry on maintenance work liable to cause ignition within the safety area.
- All equipment with operating engines within the safety area must be of an approved type.
- Nailed boots or shoes shall not be worn during the fueling operation because of the possibility of sparks.
- Fuel spillage must be prevented. The danger associated with fuel spillage is the formation of vapor; usually it is not easy to have vapors of jet A1 at ambient temperatures up to 37.8 C. **Fuel spillage must be cleaned up before engine start.** Small spills of more than 3 meters long in any direction or 4.5meter square area and of a continuing nature must be covered with foam. Ground slope or wind may carry spilled fuel to possible ignition sources. Spilled fuel shall be removed by use of absorbent or neutralizing material:
- If during fueling/defueling, the presence of fuel vapor is detected in the cabin, the flight deck and/or Ground Technician shall be informed immediately. The fueling or defueling process and all other activities using electrical equipment (e.g. cleaning) shall stop

GENERAL DUTIES AND RESPONSIBILITIES OF CABIN CREW DURING FUELING & DEFUELING (PAX ONBOARD)

- Flight deck crew informs the Team Leader that refueling is going to start and which side the refueling will be conducted
- Team Leader informs passengers & crew using the standard refueling announcement
- Team Leader shall inform all cabin crew using ALL call by intercom, “Fueling will be commenced and it will be by LT Side, RT side or both side”

All cabin crew shall be in their assigned position in the cabin and make sure both cabin and galley is ready for fueling:

- Passengers do not smoke Sign On
- **Galley shall be secured**
- All items of personal electrical devices are switched OFF
- Passengers remain seated, with their seat belts and shoulder harness (if applicable) released
- All aisles and routes to exit remain clear from obstruction (must never be blocked by unattended catering or cleaning equipment)
- **The outside area beneath each exit remains clear (if not report to the PIC)**
- The PIC shall be informed immediately if any fuel vapor is detected in the cabin
- All lavatories must be vacated and locked
- Curtains must be opened and secured
- One cabin crew member shall be positioned at each pair of aircraft doors according to the special safety regulations. (A, B, C, D) (Refer to CCOM or previous edition of FOM for the safety positions)
- Cabin crew who aren’t attending doors shall monitor the cabin until refueling is completed.



B737 WXR RADAR

Conventional Radar to IntuVue RDR-4000

Weather radar remains one of the most critical situational awareness tools available to Boeing 737 flight crews.

Used correctly, it provides early identification of hazardous weather and supports safe decision-making across all phases of flight.

Understanding not only what the radar displays, but how it generates and prioritizes information, is essential to making the most of this system.

Conventional Weather Radar Fundamentals

The Boeing 737 weather radar detects and locates precipitation-bearing clouds along the aircraft's flight path.

A forward-sweeping antenna scans an arc of 180 degrees ahead of the air-craft and presents weather returns on a color-coded display against a black background.

- **Green** indicates light precipitation
- **Amber** moderate precipitation
- **Red** heavy precipitation.

These colors represent rainfall intensity, not turbulence directly, and should always b interpreted with that limitation in mind.

In **MAP mode**, the radar can also be used for ground mapping. In this mode, highly reflective surfaces such as coastlines, terrain features, large cities, and built-up areas appear in red, amber, or green depending on reflectivity.

Ground mapping can be particularly useful in areas with limited ground-based navigation aids, but it is important to remember that the weather radar is not a proximity warning or collision avoidance system and should never be relied upon for terrain clearance.

When turbulence mode is selected, the radar displays areas of precipitation associated with significant horizontal wind flow.

Targets where precipitation movement toward or away from the antenna exceeds approximately five meters per second are displayed in magenta, indicating the potential presence of severe turbulence. Turbulence detection is automatically limited to a range of 40 nautical miles, regardless of the selected range, emphasizing its role as a short-range hazard aware-ness tool.

The IntuVue RDR-4000: A Three-Dimensional Weather Picture

Manufactured by Honeywell, the RDR-4000 IntuVue radar represents a major advancement in airborne weather detection.

Rather than relying on a single scan, the system performs multiple scans at different tilt angles and merges this information into a three-dimensional weather model. Advanced processing removes ground clutter, resulting in a clearer and more reliable picture of significant weather out to 320 nautical miles.

When operating in AUTO mode, IntuVue automatically manages tilt and gain, removing the need for manual adjustment. The system continuously scans for short, medium, and long range weather, making it effective during all phases of flight, from departure to cruise and approach.

Thunderstorm tops within 5,000 feet of the aircraft altitude are retained on the display until they no longer pose a threat, allowing crews to safely maneuver around weather that might otherwise appear less significant.

Flight Path Weather Concept

A key feature of IntuVue is the distinction between flight path weather and secondary weather. Flight path weather represents precipitation that lies directly along the aircraft's projected path, typically within ±4,000 feet vertically. At cruise altitudes above 29,000 feet, the lower boundary of this envelope may extend down to 25,000 feet to ensure relevant convective activity is displayed. During ground operations, departure, and approach, the upper limit of the envelope is fixed at 10,000 feet MSL, providing approximately ten minutes of look-ahead.

Flight path weather is displayed without markings, while secondary weather— outside the immediate flight path—is shown with black diagonal stripes. This presentation allows crews to focus on hazards that require immediate action while still maintaining broader situational awareness for strategic planning.

Manual Mode and Vertical Weather Assessment

Manual (MAN) mode provides crews with the ability to assess storm structure and vertical development. By selecting altitude "slices" from the three-dimensional weather memory, pilots can view reflectivity at constant MSL altitude levels from ground level up to 60,000 feet. These slices are corrected for the curvature of the Earth, enabling a more accurate assessment of storm height and intensity.

In MAN mode, turbulence and secondary weather information is removed to reduce clutter and enhance analysis of reflectivity. This mode is particularly useful when evaluating thunderstorm tops, building cells, or deciding whether lateral or vertical avoidance is the safer option.

Additional Hazard Detection Features

IntuVue also includes predictive hazard identification features. Predictive hail and lightning icons may appear over areas with radar signatures consistent with these hazards. While the radar does not directly detect hail or lightning, the displayed icons indicate a high probability of their presence based on data analysis.

The Rain Echo Attenuation Compensation Technique (REACT) addresses a known radar limitation: signal weakening when passing through heavy precipitation. When attenuation is significant, magenta arcs are displayed to alert the crew that weather beyond a storm cell may be more severe than shown. This feature helps prevent underestimating hazards hidden behind intense rainfall.

Predictive Windshear Protection

Predictive Windshear (PWS) is integrated into the radar system to provide advance warning of windshear ahead of the aircraft. The system operates automatically below 1,800 feet AGL, with alerts available at and below 1,200 feet AGL. Depending on the aircraft's proximity to the detected hazard, the system generates either Caution or Warning alerts, accompanied by a windshear icon on the weather display.

Thank You